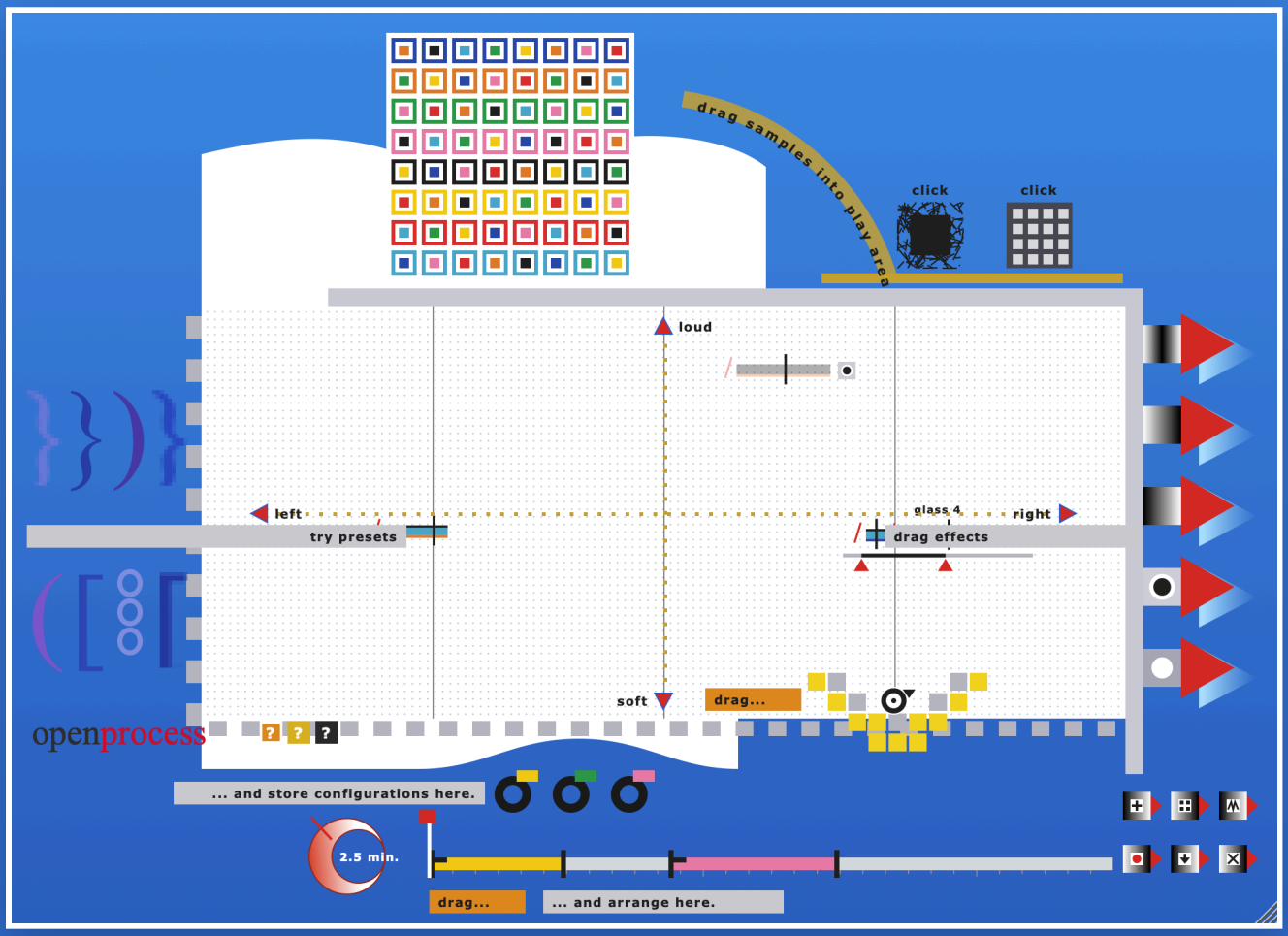
openprocess

manual · version 1.2



vst3 · vst2 · au · lv2 · standalone · macos · windows · linux

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1 about openprocess

openprocess is a loop workspace. You drag sounds into a window, they loop, and you keep changing them while they play. You store what you hear as snapshots and lay those out on a timeline. It has no notes, scales, tempo or bars.

It is modelled on **ovalprocess** (2000) by Markus Popp, who records as Oval. Programmer Richard Ross wrote it for Popp. ovalprocess was software, a series of CDs, a lecture and three public sound terminals built with the Berlin architects skotoparc. It ran on a PowerMac in an illuminated Plexiglas terminal with a trackpad, open to anyone walking past. openprocess follows the operating manual of that installation: the same screen, from top to bottom, and the same rules.

> "Ovalprocess is a model of how I work."
>
> Markus Popp, Sound On Sound, 2002

the rules it keeps

- **Everything loops.** Sounds never stop by themselves. You change their order and structure while you hear them.
- **Real time.** Every change is heard at once. The one exception is timeline playback, when nothing can be edited.
- **Top to bottom.** Pick sounds in the browser, work on them in the play area, then build a piece from snapshots on the timeline.
- **Only symbols.** No typing, no dialog boxes, no musical units. You grab, drag, drop and throw away colour-coded objects.

The original Oval sound archive is not part of openprocess. Its 64 built-in loops are synthesized and chopped like a skipping CD. Your own files can be chopped the same way (chapter 11).

2 installing and starting

openprocess comes in the **poppsphere bundle** (with opensphere and Quell): one download per system from the releases page of `github.com/femifleming/poppsphere-bundle`. On a Mac, Homebrew does it in one line: `brew install --cask femifleming/tap/openprocess`.

system	put the files here
macOS	`openprocess.vst3` → `~/Library/Audio/Plug-Ins/VST3`, `openprocess.component` → `…/Components`, `openprocess.vst` → `…/VST`, `openprocess.app` → `/Applications`. Universal: Apple silicon and Intel.
Windows	`openprocess.vst3` → `C:\Program Files\Common Files\VST3`, `openprocess.dll` (VST2) → your VST folder; `openprocess.exe` runs on its own.
Linux	`openprocess.vst3` → `~/.vst3`, `openprocess.lv2` → `~/.lv2`, `openprocess.so` (VST2) → `~/.vst`; `openprocess` runs on its own (ALSA or JACK).

In a DAW, load openprocess on an instrument track. It plays as soon as something is in the play area, whether or not the transport is running. It has nothing to do with the song tempo. It has a stereo audio input, which only *capture input* and *Input Thru* use. Hosts that offer a side-chain input for instruments can feed it there.

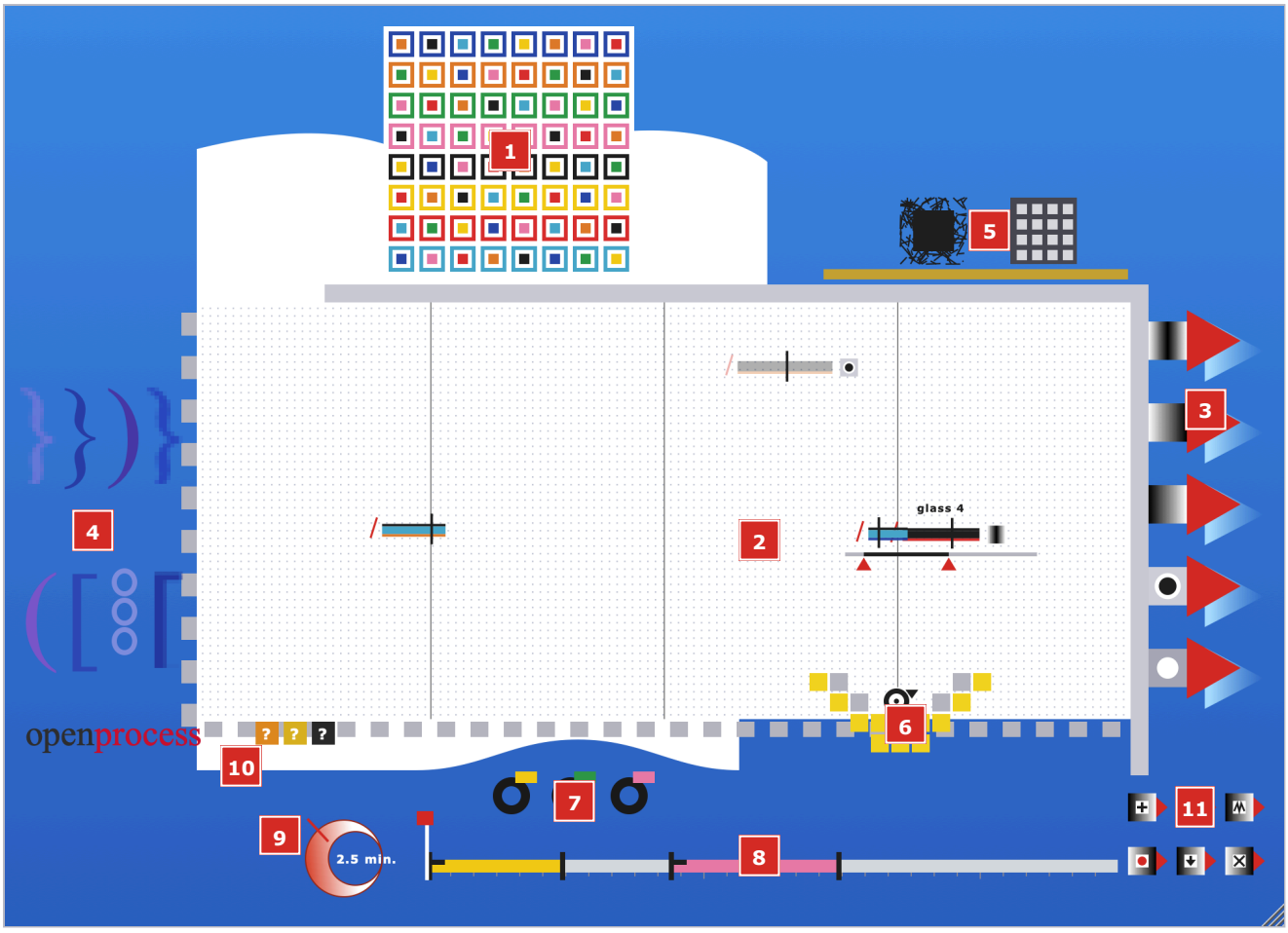
The standalone app uses the audio device set under *Options → Audio/MIDI Settings*. It asks for microphone access the first time, for the capture function. Your work is saved when you quit and is back when you start again.

The window can be resized from its lower right corner, from 600 to 1600 pixels wide. The proportions stay fixed.

quick start

1. Click a bracket on the left. Something plays.
2. Click the black scribble a few times. Loop lengths change.
3. Drag a fade onto one sound and pull another sound down to make it softer.
4. Click the yellow snapshot button. Do it again after changing things.
5. Drag both reels onto the timeline, one after the other.
6. Click the red flag.

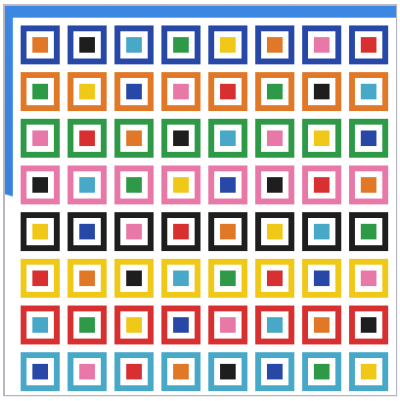
3 the screen



1 browser: 8 × 8 loops
2 play area: panorama and volume
3 effects
4 presets
5 logical operators
6 snapshot button
7 snapshots (reels)
8 timeline with play flag
9 composition length wheel
10 help marks · demo (word mark)
11 tools

Messages such as "loaded 12 files" or "the play area is full" appear for a few seconds in a grey tag at the top left.

4 the browser



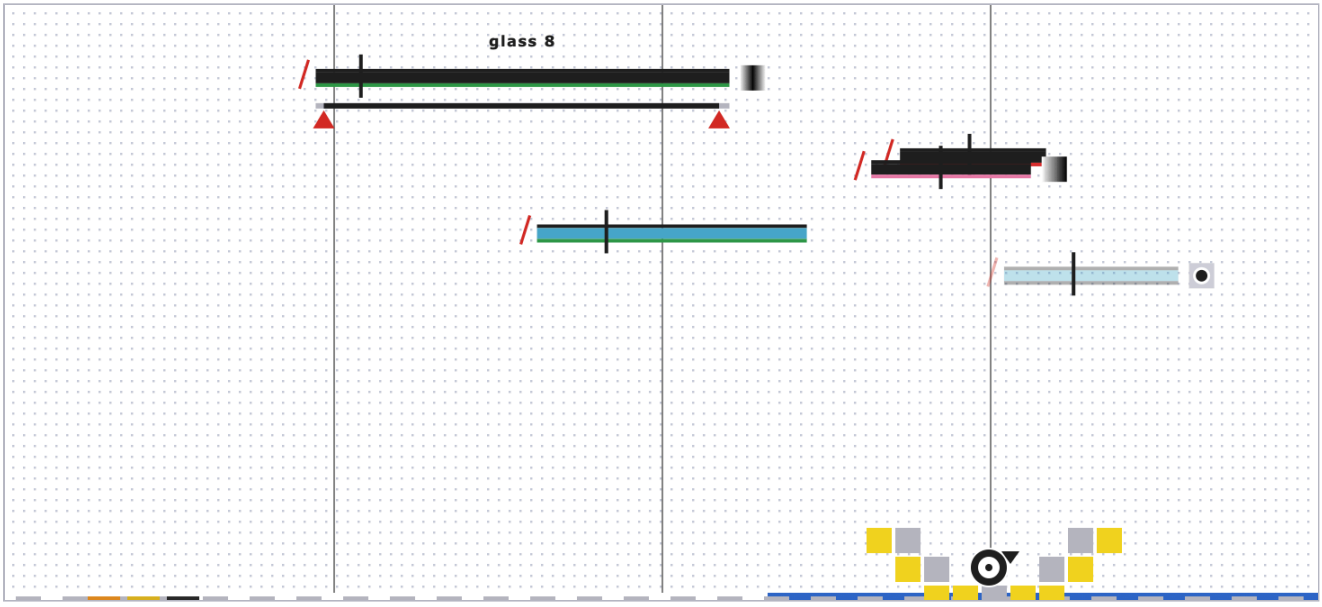
The browser holds the sound material: 64 loops in an 8 × 8 grid. The outer colour of a square shows its **row**, the family of sounds. The inner colour tells the columns apart.

Drag a square into the play area and let go. The sound starts looping at that spot. Outlined squares trail behind it as you drag. The same loop can be dragged in several times.

Click a square without dragging to select it (black frame). The *skip again* and *capture input* tools work on the selected square.

A pale square is still being made, for example just after loading files.

5 the play area



The dotted white window is where sounds play. A sound's position sets two things:

- **left ↔ right**: the stereo panorama
- **top ↔ bottom**: loud ↔ soft. The bottom edge is silent.

Each sound is drawn as a coloured bar. The bar's length shows how long its loop is. A black vertical **pointer** runs across the bar from left to right as the loop plays, again and again.

Drag a bar to move the sound. The change is heard as you move.

Throw a sound away by dragging it out of the play area. It stops.

loop length

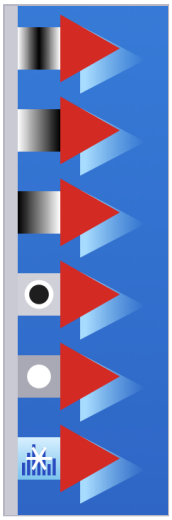
Click a bar to select it. Its name appears above it. Below it, a grey strip stands for the whole sample, with **two red pointers**. The part between them is what loops. Drag the left pointer to move the start and the right pointer to move the end. A very short loop turns any sound into a buzz or a click pattern, the classic Oval texture.

Loop ends are crossfaded over about 6 ms so that they stay smooth.

Click on an empty part of the play area to deselect.

Up to **24 sounds** can play at once.

6 effects



The six effects are on the right edge. **Drag one onto a sound** in the play area. It sticks to the sound as a small icon at the end of its bar. Each sound carries one effect at a time, and a new one replaces the old.

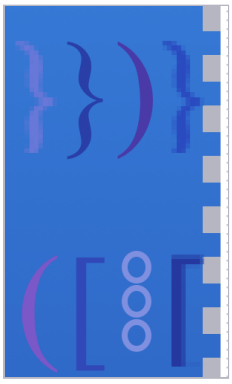
icon (top to bottom)	effect
black in the middle	**fade in / out**: volume rises, then falls, over each loop
white → black	**fade in**: rises over each loop
black → white	**fade out**: falls over each loop
black dot	**mute**: the sound keeps running, silently
white dot	**solo**: only sounds marked solo are heard
ice blue, white crystal	**freeze**: the sound becomes a held spectrum (after Jean-François Charles' freeze, as in ppooll's spectral freeze): its last 8 spectral frames are kept and every moment each frequency takes one of them at random, so it hangs in the air and shimmers

Freeze. A fraction of a second after it is dropped on, the sound freezes. **Click the icon** to freeze it again at that moment. When the sound is selected, three rails under its loop strip set **slide** (how slowly it glides into a new freeze, up to about 3 s), **denoise** (quiet frequencies drop out, leaving the strongest partials) and **again** (freeze again by itself every 0.5–20 s; at the left, only when you click). Frozen sounds are stored with these settings in snapshots. A freeze on a silent moment is silent: click again. The **matrix | scan** switch beside the rails picks the kind of freeze. **matrix** is the one above. **scan** (after Charles' freeze~) holds the last 1.4 s of frames instead and sweeps back and forth through them; the first rail then sets the **speed** (1x is as recorded, 0 holds a single frame, up to 2.8x).

Fades follow the loop, so short loops with a fade turn into pulses. Muted or soloed-out sounds are drawn faded, and their pointers keep running.

To remove an effect, drag its icon off the sound, or drop the same effect on it again. Dropping the icon on another sound moves the effect there.

7 presets

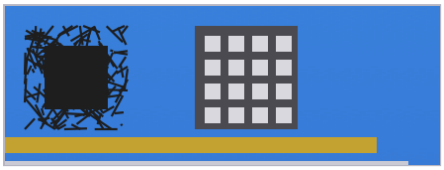


The eight brackets on the left are presets. **Click one** and the play area fills with 3 to 5 sounds, flying in from the bracket. Each bracket has a sorted pool: mostly sounds from one browser row, with some from two related rows. **Which sounds, where and how long is never the same twice.** Click again for another version.

- The **first, third, fifth and seventh** brackets use long parts of the samples, 40–100 %.
- The **second, fourth, sixth and eighth** use tiny fragments, 3–25 %.
- Each sound has a one-in-four chance of getting a fade.

A click replaces what is in the play area. **Shift-click** adds to it instead.

8 logical operators



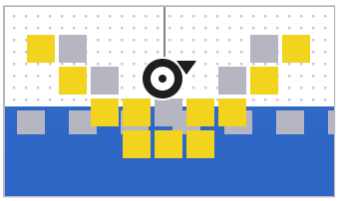
Above the play area on the right are two buttons that change everything in the play area at once, by chance.

Left (black scribble): new random loop lengths and start points for every sound, from 2 % to the whole sample.

Right (grey grid): new random positions, which is new panorama and volume for every sound. The sounds glide to their new places.

Use them as often as you like, one after the other or in turn.

9 snapshots

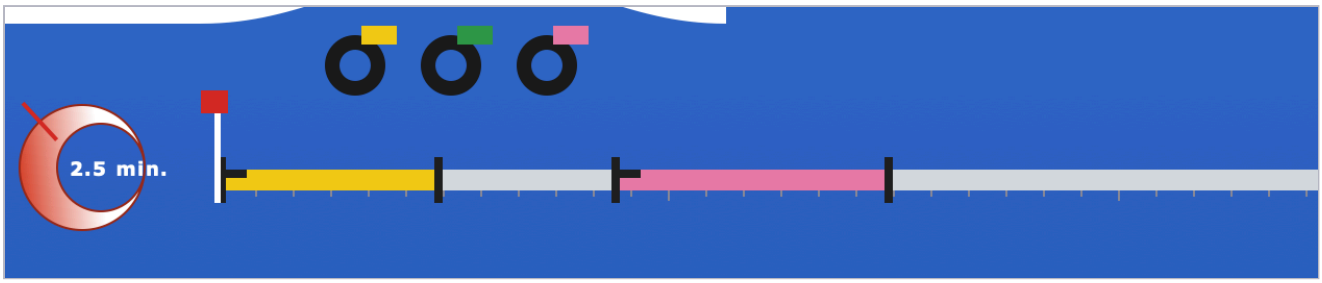


The yellow chequered shape at the bottom of the play area is the **snapshot button**. Click it and the whole play area is stored: every sound with its position, loop length and effect. The snapshot lines up below as a **reel**, a black ring with a colour tag.

- **Drag a reel onto the timeline** to use it in a composition (chapter 10).
- **Drag a reel back into the play area** to hear that configuration again. It replaces what is there.
- **Throw a reel out of the window** to delete it.

There is room for **10 reels**.

10 timeline



The timeline is the bar at the bottom. It holds the composition. Reels dropped on it become **segments** in the reel's colour: 20 seconds long, or an eighth of what the timeline shows when you are looking at a long stretch of it.

- **drag** a segment to move it
- **drag its right end** to make it longer or shorter
- **alt-drag** to copy it
- **drag it away** (up or down, off the bar) to delete it

Segments may overlap. Where they do, both play together. Each segment keeps its own copy of the snapshot, so deleting the reel does not change it. Segments fade in and out over 30 ms at their edges. There is room for **256 segments**.

wheel

The red crescent to the left sets the **length of the composition, from 1 to 30 minutes**. Drag up or right for longer, down or left for shorter. The scale is logarithmic: the first few minutes move in tenths, then halves, then whole minutes above ten.

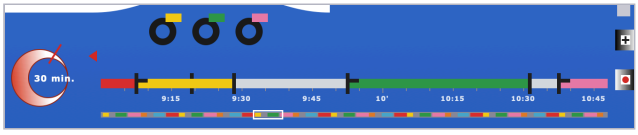
Normally segments keep their places relative to the whole, as in the original, so a longer piece stretches them too. **Hold shift while you drag the wheel** and they keep their times in seconds instead: the end of the piece moves and nothing else does (segments past a new, shorter end are dropped).

zoom

Thirty minutes do not fit on 540 pixels, so the timeline zooms. Over the timeline:

- **mouse wheel** up / away, or **pinch** on a trackpad: closer in, around the pointer
- **sideways scroll**, or **shift + wheel**: along the piece
- **double-click** the bare bar: 4× closer there, or, when zoomed, back to the whole piece

When zoomed in, a thin **overview strip** under the bar shows the whole composition with a white window round the part on screen. Drag the window, or click the strip to jump. The ticks under the bar follow the zoom, from seconds to ten minutes, with times written as `9:30` or `10'`. You can zoom in until ten seconds fill the bar.



play flag

The **red flag** on its white stem is the play pointer. **Click it** to play from the flag to the end; playback then starts over from the flag, as a loop, until you click again. **Drag the flag** to start later. When the flag is off screen a small red arrow at that end shows where it is. While playing, the view pages along with the pointer.

> While the timeline plays nothing else can be changed (zooming still works). This is the one exception to the real-time rule, as in the original.

11 sounds

the built-in bank

All 64 loops are made by openprocess itself when it starts. Each row is one family. The first three are the timbres Popp set as the rule for the *ovalprocess* CD: guitar, organ and feedback. All loops use the same set of pitches (D, E, F♯, A, B, C♯), so any of them can be combined.

row	family	what it is
1	organ	drawbar chords with a slow tremolo
2	guitar	plucked strings, loose arpeggios
3	feedback	sustained, drifting tones that swell and saturate
4	skips	clicks, blips, ticks and stuck-frame stutters
5	glass	FM bells, shimmering and decaying
6	low	soft bass swells
7	noise	swept filtered noise, hiss, crackle
8	skipped	organ and glass, heavily skipped

cd skip

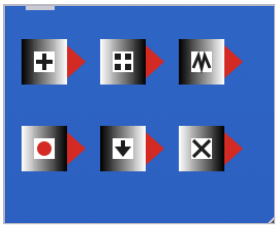
Every loop is built the way a skipping CD plays: from fragments between a few milliseconds and a second and a half long. Some fragments repeat, the read position jumps ahead or back, and now and then a "read error" adds a drop-out, a frozen bit-crushed frame or a click. **Columns run from clean on the left to heavily skipped on the right.** Loops are 0.5 to 9 seconds long and none is a multiple of another, so loops drift against each other instead of falling into a common beat.

your own sounds

- **Drop audio files or folders onto the browser** (or use the *load sounds* tool). Folders are searched with their subfolders, up to 256 files. All 64 squares are refilled. Each square gets a part of one file, chopped with the same CD-skip process, clean on the left and skipped on the right. Every file is used before any repeats. Files shorter than the square's loop length are used whole.
- **Drop a single file onto one square** to replace only that square. The first 30 seconds of the file are used unchopped.
- Formats: WAV, AIFF, FLAC, Ogg Vorbis, MP3, plus M4A/CAF on the Mac.
- **built-in sounds** (tool) brings back the original bank.

> A session remembers where its files are, not the audio itself. If files are moved, those squares keep their previous sound, and a message names the file that could not be read.

12 render, capture, tools



The six small buttons at the bottom right are additions that the installation did not have. Their names show when the mouse is over them.

button	does
+ load sounds	choose files or folders to chop into the browser
▦ built-in sounds	restore the synthesized bank
^^ skip again	re-chop the selected square with a new pattern and strength (or the square of the selected sound). A single file becomes a skipped loop.
• capture input	the last 8 seconds of audio input, silence trimmed, into the selected square (or the last square). It is saved as a WAV in `~/Music/openprocess/captures`.
↓ render to file	write a WAV file of the composition
× clear	empty the play area. **Shift-click**: clear everything (play area, reels and timeline) for a new piece

render to file

If the timeline has segments, the render runs from the play flag to the end of the composition. If not, it records the play area for the composition length set on the wheel. The file is 48 kHz, 16-bit stereo, mixed at the current Master level, and is written as it goes, so a 30-minute piece needs no more memory than a short one. The default place is `~/Music/openprocess` (your Music folder on Windows and Linux too). Rendering is faster than real time; the message in the corner counts up the percent.

13 help and demo



The three **?** marks next to the word mark open help without stopping the sound:

- **?** orange: the **guide**, the labels shown on the cover of this manual ("drag samples into play area", "loud", "soft", "try presets" …). Click it again to hide them.
- **?** yellow: a page about the play area
- **?** black: a page about snapshots, the timeline and sounds

Click the word mark to start the **demo**. A cursor works through every function on its own: it drags sounds, applies effects, uses presets and operators, takes snapshots, arranges and plays them. **Any click ends the demo** and puts back exactly what you had. As in the installation, the standalone app starts the demo by itself after three minutes without input when the play area is empty.

14 midi and parameters

note	number	action
C1 – G1	36 – 43	presets 1 – 8
G♯1	44	logical operator: lengths
A1	45	logical operator: positions
A♯1	46	snapshot
B1	47	timeline play / stop
C2 – D♯7	48 – 111	browser squares 1 – 64: add at a random place, or remove if already playing

Note names use middle C = C3 (Live, Logic). Notes act on note-on; velocity and note-off are ignored. While the timeline plays, only B1 has an effect.

parameter	range	default
Master	0 – 1	0.8
Input Thru	0 – 1	0

Both can be automated. **Input Thru** lets the audio input through to the output, for example to hear what you are about to capture. The output has a soft limiter just below full scale.

15 reference

item	value
browser	64 loops (8 rows × 8 columns), 0.5 – 9 s each
sounds in the play area	up to 24
volume	height squared; the bottom edge is silent
snapshots	up to 10
timeline segments	up to 256; new segments 20 s
composition length	1 – 30 minutes; the timeline zooms to 10 s
capture	last 8 s of input
render	WAV, 48 kHz, 16-bit, stereo, streamed to disk
user folder	`~/Music/openprocess`
saved with the session	play area, snapshots, timeline, length, flag, how each square was made, parameters
window	1000 × 720 at 100 %, resizable 60 – 160 %

sources

Ovalprocess sound installation, Dokumentation, Bedienungs- und Wartungshinweise, and Markus Popp, ovalprocess project text (Berlin, 7 April 2001), Ars Electronica archive · Markus Popp, ovalprocess US 2k lecture v0.2, Ars Electronica archive · Oval, "Ovalprocess", *Leonardo* 35/5, 2002 · Sam Inglis, "Markus Popp: Music As Software", *Sound On Sound*, October 2002 · Marc Weidenbaum, interview with Markus Popp, Disquiet / Pulse!, 1996.
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